

Experimental analysis of LoRaWAN class B synchronization: Towards NTN-integrated IoT networks

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ABSTRACT

The rapid expansion of Non-Terrestrial Networks for global IoT connectivity faces key challenges related to synchronization, latency, and energy efficiency under orbit-dependent satellite coverage. LoRaWAN Class B offers a potential solution by enabling deterministic downlink scheduling. However, its real-world performance remains largely unexplored, particularly regarding timing accuracy and energy consumption. This paper presents a comprehensive experimental evaluation of Class B synchronization accuracy and power consumption using a real testbed. Results show that absolute synchronization errors grow from 1.5 ms to 11.5 ms, while relative errors remain below 0.02%. Worst-case cumulative drift stays within the 30 ms reception window, reaching 23.6 ms. Current measurements indicate that idle consumption decreases from 16.80 mA to 6.40 mA, with Sleep mode reducing overall draw by approximately 62%. These findings provide an empirical characterization of Class B timing precision and energy efficiency, supporting the design of future hybrid terrestrial–Non-Terrestrial Networks architectures.

1. Introduction

Long Range Wide Area Network (LoRaWAN) is a prominent Low Power Wide Area Network (LPWAN) technology designed to provide long-range, energy-efficient wireless communication for Internet of Things (IoT) devices [1]. It operates in unlicensed sub-GHz Industrial, Scientific and Medical (ISM) bands using a star topology and supports transmission ranges of up to 15 km in rural and 5 km in urban environments, making it highly suitable for scalable, battery-powered applications.

The specification defines three classes (A, B, and C), each tailored to different application requirements. Class A provides the lowest energy consumption by opening receive windows only after uplinks, while Class C offers continuous reception at the expense of battery lifetime. Between these extremes, Class B introduces scheduled receive windows-called ping slots-that are synchronized to periodic beacons transmitted by the gateway (GW). This mechanism enables scheduled downlink communication while preserving energy efficiency; however, it requires precise timing synchronization, making it sensitive to configuration parameters and environmental conditions.

The scheduled downlink capability of Class B is particularly valuable in applications requiring predictable communication patterns. In smart agriculture, for instance, it enables energy-efficient connectivity for irrigation control, soil monitoring, and climate-aware decision-making [2]. Similar requirements exist in logistics, environmental monitoring, and smart-city infrastructures, where large-scale IoT deployments must operate autonomously for extended periods while maintaining reliable downlink connectivity for configuration updates and command delivery.

However, terrestrial LPWAN coverage often proves insufficient in remote or geographically challenging areas. Satellite-enabled LoRaWAN, particularly through Non-Terrestrial Networks (NTNs), represents a promising alternative by providing global coverage [3]. The integration of Class B with satellite infrastructure introduces significant challenges, including maintaining reliable synchronization across long distances, mitigating timing errors exacerbated by satellite mobility, and optimizing energy consumption under constrained link budgets and strict duty-cycle regulations.

Despite its potential, Class B remains one of the least explored LoRaWAN classes in both terrestrial and non-terrestrial deployments.

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List of Acronyms			
Acronym	Definition		
5G	Fifth Generation	LR-FHSS	Long Range Frequency Hopping Spread Spectrum
6G	Sixth Generation	MAC	Medium Access Control
6G-LoRaGRAN	6G Long Range Gateway Radio Access Network	M/G/1	Markovian/General/1 Queueing Model
ABP	Activation By Personalization	MC2AMB	Multi-Class Multi-Channel ALOHA-based Model
ADR	Adaptive Data Rate	MCU	Microcontroller Unit
AES	Advanced Encryption Standard	mMTC	massive Machine-Type Communication
AI	Artificial Intelligence	NLOS	Non-Line-of-Sight
AS	Application Server	NOMA	Non-Orthogonal Multiple Access
BW	Bandwidth	NS	Network Server
CR	Coding Rate	NS-3	Network Simulator 3
CRC	Cyclic Redundancy Check	NTN	Non-Terrestrial Network
CSS	Chirp Spread Spectrum	OTAA	Over-The-Air Activation
DR	Data Rate	PDR	Packet Delivery Ratio
DRL	Deep Reinforcement Learning	PPM	Parts Per Million
DtS-IoT	Direct-to-Satellite Internet of Things	PPS	Pulse Per Second
ECC	Electronic Communications Committee	RC	Resistor-Capacitor
ED	End Device	RF	Radio Frequency
GEO	Geostationary Earth Orbit	RSSI	Received Signal Strength Indicator
GPIO	General Purpose Input/Output	RTC	Real-Time Clock
GPS	Global Positioning System	RX	Reception
GW	Gateway	SDN	Software-Defined Networking
HAP	High Altitude Platform	SF	Spreading Factor
HSI	High-Speed Internal	SNR	Signal-to-Noise Ratio
IoT	Internet of Things	SPI	Serial Peripheral Interface
IP	Internet Protocol	SRD	Short-Range Device
ISM	Industrial, Scientific and Medical	TCXO	Temperature Compensated Crystal Oscillator
LEO	Low Earth Orbit	TDMA	Time Division Multiple Access
LLN	Low-power and Lossy Networks	TX	Transmission
LoRaWAN	Long Range Wide Area Network	UAV	Unmanned Aerial Vehicle
LPWAN	Low Power Wide Area Network	UDP	User Datagram Protocol
		UNB	Ultra-Narrowband
		VBS	Virtual Basic Station

While simulation-based studies have investigated its theoretical performance, there exists a critical gap in experimental validation using real hardware. This lack of empirical evidence is particularly striking given the growing demand for predictable and energy-efficient downlink connectivity in large-scale IoT systems.

This paper addresses this research gap by presenting the design, implementation, and evaluation of a Class B LoRaWAN testbed based on commercially available STM32 microcontroller Units (MCUs) with SX1276 transceivers. Through controlled experiments, we measured timing precision across multiple ping slot configurations, analyzed synchronization errors and their accumulation patterns, and characterized energy consumption profiles. Our work provides the first comprehensive experimental analysis of Class B behavior under realistic conditions, quantifying fundamental trade-offs between responsiveness, synchronization accuracy, and power efficiency.

Furthermore, this investigation establishes a foundation for future integration of terrestrial LoRaWAN with emerging network architectures. As Sixth Generation (6G) systems envision seamless terrestrial-NTN convergence to achieve universal coverage, understanding and optimizing the synchronization mechanisms of Class B operation becomes increasingly relevant. The insights from our experimental testbed contribute to this vision by identifying performance bottlenecks and configuration guidelines that could inform the design of massive IoT deployments in next-generation networks.

2. LoRaWAN fundamentals

A LoRaWAN network is typically organized into four core elements, according to the LoRaWAN specification [1]: End Devices (EDs), GWs,

the Network Server (NS), and Application Servers (AS). EDs are responsible for sensing and transmitting data, GWs forward messages between EDs and the NS, the NS manages the network, including device registration and security, and the AS processes application-specific data.

At the physical layer, LoRaWAN relies on LoRa modulation, a proprietary Chirp Spread Spectrum (CSS) technique. LoRa encodes information in chirped signals whose frequency increases or decreases linearly over time. Each transmission consists of a preamble, which allows the receiver to detect the signal and synchronize, followed by the payload containing the actual data. Key parameters include the Spreading Factor (SF), Bandwidth (BW), and Coding Rate (CR). The SF determines the number of symbols per bit, affecting transmission time, range, and sensitivity. Higher SFs provide longer range and better robustness at the cost of lower Data Rates (DR), while lower SFs enable higher throughput but shorter range. The BW defines the frequency range of the chirp, influencing both DR and resilience to interference. The CR adds forward error correction to enhance reliability in noisy channels.

To optimize both performance and energy efficiency, LoRaWAN defines a set of Medium Access Control (MAC) commands for network management, including device configuration and an Adaptive Data Rate (ADR) mechanism that dynamically adjusts SF and transmit power for each device based on link quality. Devices operate in unlicensed ISM bands, subject to regulatory limitations such as duty cycle restrictions [4]. For example, in the 868 MHz band, a 1% duty cycle restricts transmission to a maximum of 36 s per hour, which impacts downlink capacity and scheduling strategies, particularly for Class B and Class C devices.

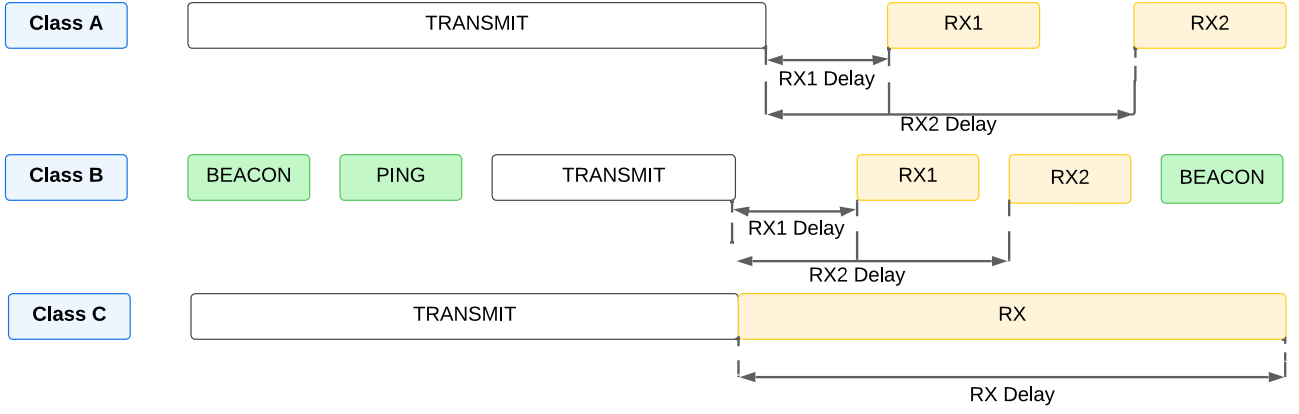


Fig. 1. LoRaWAN device classes.

2.1. LoRaWAN device classes

LoRaWAN defines three ED classes (Fig. 1), offering distinct energy-downlink latency trade-offs:

- **Class A (Mandatory):** Employs an ALOHA-based approach. Each uplink opens two short, sequential downlink windows (RX1, RX2). Optimized for minimal energy use; downlinks are only possible after an uplink. Ideal for battery-powered sensors with infrequent uplinks.
- **Class B (Beacon-Synchronized):** Extends Class A by adding scheduled, periodic downlink reception windows (*ping slots*) synchronized via GWs beacons. Enables deterministic downlinks crucial for latency-sensitive applications (e.g., irrigation control). Energy consumption is moderate, higher than Class A but significantly lower than Class C, due to periodic receiver activation.
- **Class C (Continuous Reception):** Keeps the receiver open almost continuously (except during transmission). While this minimizes downlink latency, the resulting high energy consumption restricts its applicability to devices with a permanent power supply.

2.2. LoRaWAN class B operation

Class B enables scheduled downlink communication by synchronizing EDs through Global Positioning System (GPS) synchronized beaconing. This mechanism provides deterministic downlink opportunities, addressing the limitation of Class A where downlinks can only occur after an uplink transmission. Fig. 2 illustrates the operational timeline, comprising several key phases: network joining, initial time synchronization, ping slot configuration, beacon acquisition and clock adjustment, scheduled ping slot operation, and normal uplink/downlink activity.

The ED initially joins the network in Class A mode, either through Over-The-Air Activation (OTAA) or Activation By Personalization (ABP), by sending a `JoinRequest`. Upon reception, the NS responds with a `JoinAccept`, establishing security parameters and enabling bidirectional communication. Subsequently, the ED requests a time reference using `DeviceTimeReq`; the NS replies with a GPS-derived timestamp (`DeviceTimeAns`), providing an initial, approximate network-aligned clock.

Once synchronized, the NS configures the ED's ping slot periodicity and downlink channel via `PingSlotInfoReq` and `PingSlotChannelReq`. Periodicity values range from 0 to 7, yielding slot intervals of $T_{\text{ping}} = 0.96 \times 2^{\text{Periodicity}}$ s. Every 128 s, GWs transmit GPS-synchronized beacons that allow the ED to switch to Class B, aligning its internal clock to compensate for drift and propagation delays.

During each beacon period, the ED opens short reception windows of 30 ms at calculated times:

$$T_{\text{slot}} = T_{\text{reserved}} + \left[\text{pingOffset} + N \times \frac{4096}{\text{pingNb}} \right] \times 30 \text{ ms},$$

where `pingOffset` is a pseudo-random offset computed via Advanced Encryption Standard (AES):

$$\text{pingOffset} = \text{AES}_{016}(\text{beaconTime} \parallel \text{DevAddr}) \bmod \left(\frac{4096}{\text{pingNb}} \right).$$

This mechanism ensures temporal diversity while maintaining predictable intervals, essential for duty-cycled operation. Downlinks are processed during these ping slots, while uplinks follow the standard Class A procedure with RX1/RX2 windows.

Each beacon frame includes a 4-byte GPS-based Unix timestamp and a 2-byte Cyclic Redundancy Check (CRC-16) for integrity verification. Optional fields may carry GW-specific metadata, such as geographic location or an identifier, using `InfoDesc` values from 0 to 3.

Beacons are transmitted every 128 s, starting from the GPS epoch. The exact transmission time of a beacon, B_T , is given by:

$$B_T = k \cdot 128 + T_{\text{BeaconDelay}},$$

where k is the smallest integer such that $k \cdot 128 > T$, and T is the number of seconds elapsed since the GPS epoch. The term $T_{\text{BeaconDelay}}$ (typically $1.5 \text{ ms} \pm 1 \mu\text{s}$) accounts for the time required by the GW's radio to switch from receive to transmit mode [1].

This timing is critical for Class B operation, as EDs use the beacon transmission start time as a synchronization reference to open ping slots at regular intervals. The NS must therefore take $T_{\text{BeaconDelay}}$ into account when scheduling Class B downlinks. Since GPS time is strictly monotonically increasing and unaffected by leap seconds, it ensures stable and consistent timing across the network.

If a beacon is lost, the ED progressively increases its listening window to compensate for clock drift, maintaining Class B operation for up to 120 min without resynchronization. After receiving a new beacon, slot timing is reset; exceeding 120 min without reception reverts the device to Class A.

The fixed beacon period ($T_{\text{beacon}} = 128 \text{ s}$) is subdivided into reserved, guard and usable intervals. Given a reserved time of 2.120 s and a guard time of 3.000 s [1], the usable window for ping slots is:

$$T_{\text{usable}} = T_{\text{beacon}} - T_{\text{reserved}} - T_{\text{guard}} = 122.880 \text{ s}.$$

It is important to note that the presence of T_{reserved} and T_{guard} introduces deterministic idle intervals in the timeline, but these are protocol-defined structural parameters known by both the NS and the ED. Therefore, unlike clock drift or jitter, these intervals do not contribute to synchronization error. The synchronization analysis presented in the subsequent sections focuses exclusively on unintended deviations caused by hardware imperfections (e.g., oscillator instability) and processing latencies.

Mobility is facilitated by the optional `GwSpecific` field in beacons, which allows EDs to identify the GW source using either GPS coordinates or a `NetID+GatewayID` pair (`InfoDesc` 0–3). When a change of serving

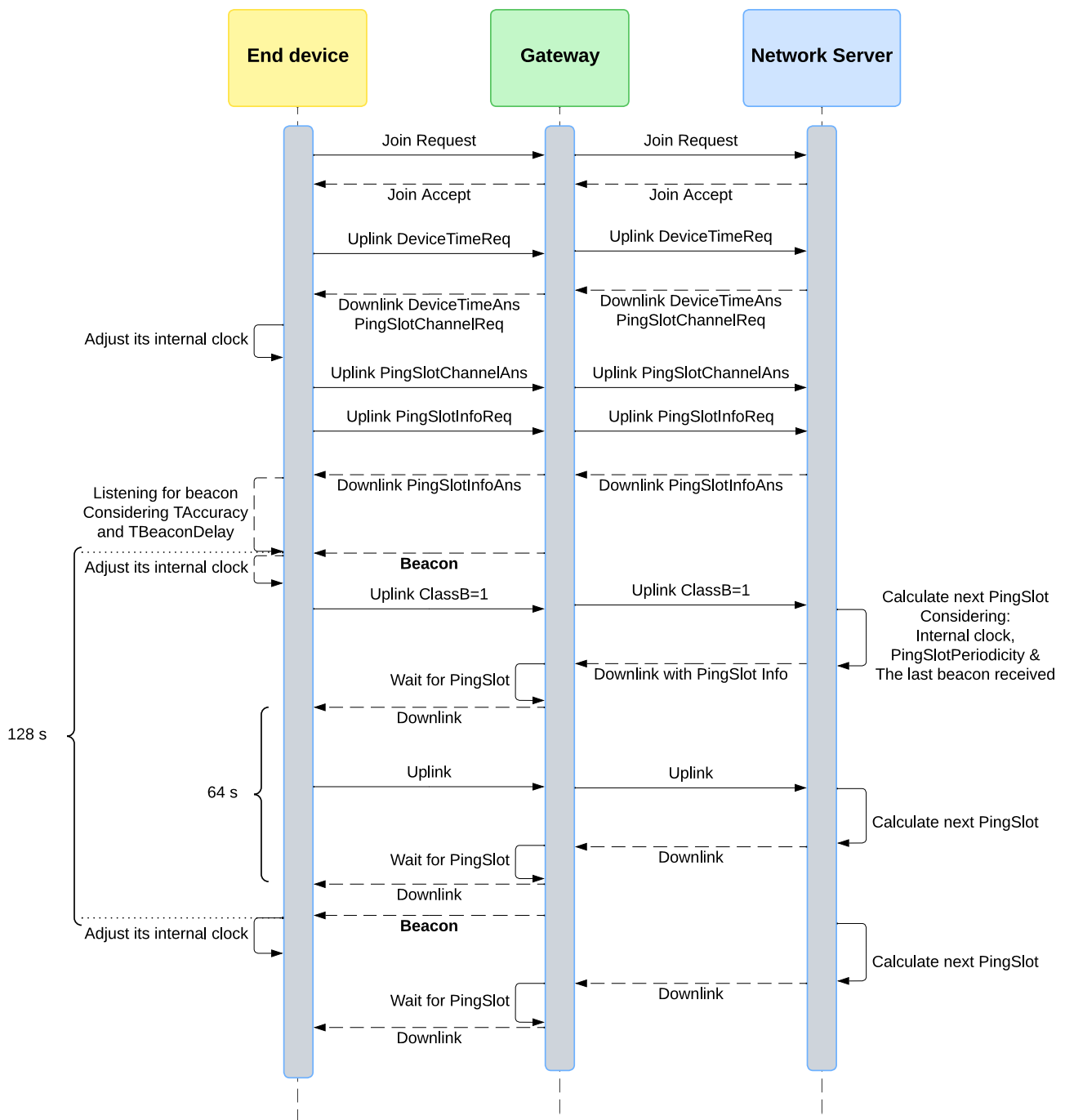


Fig. 2. LoRaWAN class B operation timeline for PS64 configuration [1].

GW is detected, EDs usually notify the NS by sending a Class A uplink, thereby updating the downlink routing path. To avoid uplink congestion during handovers, a random delay of up to 120 s is introduced before resuming transmissions. For devices with limited mobility, periodic uplinks without decoding beacon metadata offer a simpler alternative for route maintenance [1].

3. Related work

A review of the literature reveals significant efforts in Class B optimization, yet critical gaps persist in experimental validation. This section presents an overview of the most relevant contributions related to

the main topics addressed in this work in order to understand the findings and outcomes.

3.1. Class B optimization and simulation studies

Ron et al. [5] developed an analytical Markovian/General/1 Queueing Model (M/G/1) for downlink optimization. By simulating terrestrial networks with varying traffic density (ρ) and SF6–SF12, they demonstrated trade-offs between delay reduction (up to 64%) and energy consumption through ping slot parameter (k) optimization. However, their work assumed ideal synchronization and GW conditions, omitting real-world radio irregularities.

Elbsir et al. [6] conducted Network Simulator 3 (NS-3) simulations comparing Classes A and B. Their results showed that Class B reduces downlink access delay from 923 s to 52 s and packet loss from 42% to 26%. Crucially, they identified collision sensitivity: higher ping slot counts (PS4–PS32) reduced latency but increased packet loss by 18% in congested networks. Limitations included fixed SFs and no ADR implementation.

Addressing ADR, Todoli-Ferrandis et al. [7] proposed a bidirectional mechanism for industrial Class B networks. Their NS-3 simulations showed 22% lower packet loss in factory deployments but relied on idealized channel models without hardware validation.

Expanding the scope of simulation analysis, Jagatha et al. [8] explored Class B and C devices through a systematic NS-3 simulation framework. They found that confirmed traffic improves with more transmission attempts and parallel receive paths, while unconfirmed traffic is largely unaffected.

Complementing these studies, Finnegan et al. [9] presented the first NS-3 implementation of LoRaWAN Class B. Their simulations revealed that GW duty cycle limits are the primary bottleneck for Class B downlink scalability, with only 64 EDs supported at the lowest DR while maintaining a Packet Delivery Ratio (PDR) above 95%. Moreover, they uncovered a protocol-level flaw: under certain configurations, a GW may transmit downlink frames that block its own subsequent beacon transmissions, disrupting network-wide synchronization.

Finally, Elbsir et al. [10] created a custom NS-3 module optimizing ping slots under duty cycle constraints. While their dynamic retransmission strategy reduced losses by 30%, all these studies, however, are limited to simulations without experimental validation.

3.2. Energy consumption studies

In the domain of energy consumption analysis, Cheong et al. [11] conducted one of the first experimental comparisons of LoRaWAN classes, focusing on power consumption. Results showed that Class A EDs can reach a 10-year battery life with proper payload, transmission interval, and SF settings, whereas Class C EDs are unsuitable due to continuous reception. However, their work explicitly excluded Class B due to unavailability of a driver implementation at the time, leaving a gap in experimental validation of Class B power consumption.

Building upon this foundation, Elbsir et al. [6] conducted a comprehensive NS-3 simulation study evaluating Class B performance for actuator-based IoT applications. The results demonstrated that higher ping slot counts can reduce link access delay under low network loads but increase energy consumption and collision probability in congested scenarios. Notably, they identified that duty cycle limitations at the GW account for approximately 55% of downlink packet losses, while RX2 window conflicts contribute 33% and internal ping slot collisions 12%. Their optimization analysis revealed that disabling RX2 reception for Class B EDs can improve downlink PDR from 38% to 82% in realistic scenarios. Despite offering insights into Class B energy-performance trade-offs, the study remains simulation-only, without experimental power measurements. To the best of our knowledge, no subsequent studies have comprehensively analyzed the power consumption of Class B devices through experimental measurements across different processor modes and ping slot configurations.

3.3. Non-terrestrial networks and satellite IoT

Beyond terrestrial networks, Direct-to-Satellite Internet of Things (DTS-IoT) has emerged as a complementary paradigm to provide global coverage where terrestrial connectivity is limited or unavailable [3,12]. Early work by Qu et al. [13] proposed a foundational LoRaWAN-based architecture for Low Earth Orbit (LEO) satellite IoT networks, defining device awakening, seeking, and store-forward versus on-board processing modes. Their framework showed the potential of LEO satellites as LoRaWAN GWs but lacked implementation or performance evaluation.

Ullah et al. [12] studied massive Machine-Type Communication (mMTC) with LEO satellites, using LoRaWAN sensors for remote connectivity. Their work presents both indirect (device-GW-satellite) and direct (device-satellite) implementation strategies and evaluates the feasibility of uplink communication from terrestrial LoRaWAN devices to LEO satellites via Monte Carlo simulations. Key findings include the influence of SF selection, sensor density, and satellite mobility on packet success probability.

Building upon this foundation, Ullah et al. [14] further explored LoRaWAN and LEO satellite integration for offshore wind farm monitoring. The results revealed that while direct satellite connectivity outperforms indirect approaches for SF7 (32.27% higher packet delivery), indirect architectures with local GWs achieve better performance for higher SFs. However, both approaches face significant challenges: indirect architectures suffer from Non-Line-of-Sight (NLOS) propagation due to turbine shadowing, while direct architectures contend with co-SF interference and limited channel budget variation that diminishes capture effect benefits.

More recently, Novero et al. [15] proposed a LoRaWAN-LEO integration using a Virtual Basic Station (VBS) for protocol transparency without changes to EDs or satellites. Operating in unlicensed bands and leveraging store-and-forward paradigms, their solution addresses scalability through Kubernetes-based orchestration while maintaining cost efficiency. However, their implementation remains limited to uplink-only communication and relies on simulated NTN paths, leaving practical validation of the complete satellite-terrestrial integration as future work.

Giambene et al. [16] investigated a hybrid NTN architecture combining Unmanned Aerial Vehicles (UAVs) with satellite backhaul for remote area monitoring. They proposed a modified Class B protocol with beacon coordination and used simulations to jointly optimize UAV cruise speed and LoRa parameters under energy constraints, achieving efficient alarm detection for agricultural and environmental monitoring applications. However, like other satellite IoT studies, their approach relies on theoretical models and simulations without experimental validation of the proposed modified Class B mechanism.

Chen et al. [17] provide a broader overview of Satellite IoT networks, highlighting challenges such as massive connectivity, low power, high mobility, stringent delay, and wide coverage, and discuss potential solutions including grant-free access, Non-Orthogonal Multiple Access (NOMA), multi-LEO cooperation, and integration with terrestrial IoT. Their analysis underscores that satellite IoT can complement terrestrial Class B LoRaWAN deployments, particularly in remote or unpopulated areas, while revealing the critical design trade-offs inherent in satellite-based massive IoT systems.

Expanding the scope to underground and complex environments, Lin and Alouini [18] conducted a comprehensive connectivity analysis of LoRaWAN-based NTNs for subterranean mMTC. Their results show that Long Range Frequency Hopping Spread Spectrum (LR-FHSS) outperforms for High Altitude Platform (HAP) and LEO satellite links due to robustness against interference, while LoRa SF7 is effective for short-range UAV communications. However, like other NTN studies, their work remains simulation-based and does not address the experimental validation of Class B synchronization mechanisms in such challenging environments.

3.4. Experimental validation and real deployments

In the domain of experimental validation, Tang et al. [19] made a significant contribution by conducting comprehensive measurements of LoRa signal propagation in urban areas using both aerial and ground GWs. Their dataset of 28,576 raw entries provides valuable real-world measurements of Received Signal Strength Indicator (RSSI), Signal-to-Noise Ratio (SNR), and high-precision timestamps. Results indicate that a single aerial GW at 150 m altitude could achieve coverage equivalent to at least three ground GWs in a 2.6 km² urban area, with significant

improvements in PDR for higher-floor devices. However, this study focuses exclusively on uplink communication and does not investigate Class B synchronization mechanisms or downlink performance. Complementing large-scale deployments, Bäumker et al. [20] experimentally analyzed power consumption in LoRaWAN nodes using SX1276 and SX1262 transceivers. Optimized module selection (SX1262) and voltage reduction achieved sleep currents as low as $0.89\ \mu\text{A}$ and transmission currents of $4.5\ \text{mA}$ at $3\ \text{dBm}$ output power. Software optimizations cut communication energy overhead by over 98% (from $34\ \mu\text{J}$ to $0.4\ \mu\text{J}$). The study provides reference values for ultra-low-power design but does not address Class B synchronization.

Most recently, Rolland et al. [21] specifically addressed Class B beacon synchronization for Direct-to-Satellite LoRaWAN, proposing a “listen-first” wake-up strategy using ABP to reduce energy consumption. Their simulation-based study demonstrated that optimal configuration of sleeping time intervals can achieve 33% energy reduction at the cost of slightly increased beacon acquisition time (from 10.7 to 13.7 h) in single-satellite coverage scenarios. This work represents one of the few attempts to leverage Class B beacons for satellite synchronization, though it remains constrained to simulation environments and does not validate the proposed wake-up strategy on physical hardware or address the practical challenges of beacon reception in dynamic satellite passes.

In a complementary direction, Scalabrini et al. [22] proposed a lightweight adaptive mechanism to improve synchronization and energy efficiency in terrestrial LoRaWAN Class B networks. Their approach predicts clock drift based on temperature and dynamically adjusts the beacon period. Implemented on commercial hardware with only 2.5 kB of memory, it significantly reduced the inter-beacon synchronization error and tripled battery lifetime compared to standard Class B configurations. Unlike simulation-based studies, this work includes extensive experimental validation in both controlled and outdoor environments, demonstrating its practical applicability for synchronized IoT applications such as structural health monitoring.

From a modeling perspective, Lamdani et al. [23] proposed a novel Multi-Class Multi-Channel ALOHA-based Model (MC2AMB) for accurate LR-FHSS performance evaluation in Dts-IoT scenarios. The model enables optimization of key LR-FHSS parameters such as header replica count and payload fragmentation, demonstrating that the industry-standard configuration of three header replicas for coding rate $1/3$ indeed represents the optimal trade-off. However, like other analytical works, their validation relies on simulation rather than experimental testing, and the model does not incorporate the specific timing constraints of Class B beacon synchronization that are crucial for efficient satellite pass utilization.

In a previous work by the authors [24], one of the first experimental validations of a synchronization and Time Division Multiple Access (TDMA) scheduling scheme over LoRaWAN Class A was presented. STM32L0 clock drift was measured in a temperature-controlled chamber, achieving a mean synchronization accuracy of 10 ms, and it was shown that synchronization can improve the PDR by up to 29% under high network loads compared to ALOHA-based setups. Although the study focused on Class A devices and a custom scheduling mechanism, it provides a valuable experimental baseline for timing accuracy in LoRaWAN. The identified research challenges and future directions — including dynamic clock skew measurement, beacon-based synchronization, and the influence of the clock source — directly motivated the present work.

3.5. Future integrated architectures and 6G perspectives

Looking toward future network architectures, Varma et al. [25] proposed a LoRaWAN scheduling mechanism tailored for 6G-based LEO satellite communications. The mechanism achieves a PDR of approximately 90% in networks scaling up to 1000 EDs, while maintaining reasonable power consumption and latency. It also supports urgent communications through dedicated channels and incorporates topology

adaptation capabilities. Although this work represents an important step toward 6G-integrated LoRaWAN systems, it remains simulation-based and has not yet been validated on hardware.

Building on this foundation, Suomalainen et al. [26] provide a comprehensive cybersecurity analysis for tactical 6G networks, highlighting the critical role of integrated terrestrial and non-terrestrial connectivity for mission-critical communications. Their work emphasizes that 6G networks will require fully integrated NTN capabilities to provide global coverage for public safety operations, with Artificial Intelligence (AI) security controls and autonomous network management becoming essential components. While their focus is on broadband mission-critical applications, the architectural principles and security considerations are highly relevant for LoRaWAN-based NTN deployments, particularly regarding the synchronization mechanisms and reliability requirements that Class B operation could provide for massive IoT scenarios in 6G contexts.

Looking toward integrated terrestrial-satellite architectures, Chinchilla-Romero et al. [27] recently introduced 6G Long Range Gateway Radio Access Network (6G-LoRaGRAN), a comprehensive testbed framework for researching LoRaWAN and Fifth Generation (5G) integration. The testbed implements network slicing at both LoRaWAN and 5G levels, enabling channel frequency reservation and radio resource isolation through dedicated slices. Two key experiments demonstrated the platform’s capabilities: first, validation against theoretical collision probabilities under varying channel configurations and traffic patterns; second, confirmation of effective inter-slice isolation between two LoRaWAN slices with different channel masks and traffic characteristics. While their work focuses primarily on terrestrial 5G integration and does not address Class B functionality or satellite connectivity, it provides a valuable architectural blueprint for future LoRaWAN-NTN testbeds and demonstrates the feasibility of large-scale, remotely accessible experimental platforms for LoRaWAN research.

In parallel with these architectural advancements, Software-Defined Networking (SDN) is gaining traction as a paradigm for flexible IoT management. Recent works have explored SDN-based LoRaWAN for GW selection and resource allocation [28], and even for underwater sensor networks where LoRa serves as a dedicated control plane [29]. Class B represents a promising candidate for this control plane due to its scheduled ping slots.

In this direction, Ayofe et al. [30] recently proposed an SDN-based architecture for NTNs, focusing on enhancing resilience through distributed controllers and intelligent traffic steering. However, despite such advancements, there is still a significant research gap regarding the specific convergence of SDN, LoRaWAN, and NTN. As noted in the survey by Djennadi et al. [31], while SDN has been widely investigated for terrestrial Low-power and Lossy Networks (LLNs), its application to satellite-integrated LoRaWAN ecosystems remains largely underexplored. This highlights the need for specialized management frameworks capable of handling the unique propagation delays and mobility patterns of space-based assets within a software-defined LoRaWAN framework.

3.6. Summary and research gap

While prior studies offer valuable insights into Class B performance, there is an almost universal dependence on simulation environments. This methodological gap severely limits the practical applicability of their findings. The existing literature primarily focuses on latency and energy optimization but fails to experimentally address the timing and synchronization constraints inherent to Class B operation. In the realm of NTN, research often remains theoretical or simulation-based, without experimental validation of Class B functionality. Crucially, to the best of our knowledge, no previous study has conducted an experimental evaluation of Class B under realistic terrestrial conditions. This omission is particularly significant for NTN scenarios, where Class B beacon synchronization could be leveraged for efficient coordination during

Table 1
Summary of experimental and simulation studies in related work.

Ref.	Methodology	Scope	Class B	EDs	GWs	Parameters Evaluated
[5]	Simulation	LoRaWAN	✓	1k	1	Avg. Waiting Time, Avg. Frames in Queue, Avg. Energy Consumption, Throughput
[6]	Simulation & Analysis	LoRaWAN	✓	4k	1	Packet Loss, Link Access Delay
[7]	Simulation	LoRaWAN	✓	200	1	PDR, Energy Consumption, SF Evolution, Impact of SNR Sample Counter
[8]	Simulation	LoRaWAN	✓	N/A	1	UL-PDR, CPSR, TX Attempts, Packet Loss Causes, GW Parallel RX Paths
[9]	Simulation	LoRaWAN	✓	> 1k	1	PDR, Beacon Delivery Ratio, ED Scalability, DC Impact
[10]	Simulation	LoRaWAN	✓	100-5k	1	Packet Loss, Link Access Delay
[11]	Hardware	LoRaWAN	✗	1	1	Current Consumption, Battery Life, SF, Payload Size
[12]	Simulation	LoRaWAN with LEO	✗	100k-500k	1	Success Probability, Collision Rate, SF, Distance, Number of EDs, Capture Effect
[14]	Simulation	LoRaWAN with LEO	✗	8.1k	1	Network Architecture, Rician Fading, PDR, SF, LOS/NLOS, co-SF Interference
[16]	Simulation	UAV-NTN	✓	2.5k	1	Detection Probability, Alarm Delay, UAV Speed, SF, Energy Constraints
[18]	Simulation	LoRa & LR-FHSS	✗	> 100k	1	Success Probability, Burial depth, NTN Platforms, Environment, Modulation
[19]	Hardware	LoRaWAN	✗	28	4	RSSI, SNR, PDR, GW Altitude, ED Location, Distance, Environment
[20]	Hardware	LoRaWAN	✗	1	1	Current consumption, SF, output power, module comparison (SX1276, SX1262), supply voltage impact, software optimization
[21]	Simulation	LoRaWAN with LEO	✓	1	1	First Catch Time, Energy consumption, Sleeping Time, Reception Window
[22]	Hardware	LoRaWAN	✓	4	1	Sync error, battery lifetime, beacon period, clock drift, temperature thresholds, step and accumulated error
[23]	Simulation	LR-FHSS	✗	> 100k	1	Packet success probability, Fragmentation, Header replica count, Traffic density
[24]	Hardware	LoRaWAN	✗	4	1	Synchronization Accuracy, Clock Skew/Drift, PDR
[25]	Simulation	LoRaWAN for 6G-LEO	✗	1k	2	PDR, Power consumption, Schedulability, Channels, Latency, Number of EDs
[27]	Hardware	LoRaWAN	✗	104	11	Collision probability, Inter-slice isolation, Network Slicing configuration, Traffic patterns

satellite pass windows, and becomes increasingly relevant in the context of emerging 6G architectures. Consequently, the practical behavior and performance of Class B mechanisms remain largely unexplored. This study fills a critical gap by providing the first empirical characterization of Class B synchronization under realistic conditions. Our findings establish a hardware performance baseline that is currently missing in the literature, and highlight the importance of experimental data and validation to enable accurate modeling of timing uncertainties in real-world EDs. Such empirical foundations are essential for the development of advanced network architectures, including SDN-based systems and 6G-integrated NTN. By quantifying these constraints, we provide key parameters that can support the design of reliable orchestration and scheduling algorithms for future large-scale deployments.

Table 1 provides an overview of the reviewed literature, highlighting differences in methodology, scope, Class B involvement, number of EDs and GWs, and the key performance metrics assessed.

Furthermore, to enable a direct technical comparison of methodological approaches and evaluated parameters, Table 2 presents a focused analysis comparing this work with a subset of the most relevant studies. From a synchronization perspective, Scalabrín et al. [22] reported a synchronization error of 610 μ s over a 30-min period in Class B hardware measurements. Garrido-Hidalgo et al. [24] experimentally validated a synchronization scheme for Class A, measuring a clock skew leading to a 375 ms drift in 60 min and a mean synchronization error of 10 ms, improving PDR by up to 29% compared to Aloha. Likewise, Cheong et al. [11] reported hardware measurements for Classes A and C, confirming the infeasibility of Class C for battery-powered devices and measuring a TX current of 95 mA, lower than the datasheet value of 117 mA. In terms of power consumption baselines, Bäumker et al. [20] experimentally demonstrated that optimized hardware selection and voltage reduction can achieve sleep currents as low as 0.89 μ A and transmission currents of 4.5 mA, providing reference values for ultra-

low-power LoRaWAN node design. This comparison emphasizes that, unlike previous works, our study provides a comprehensive experimental evaluation of Class B devices, addressing timing accuracy, power consumption, and synchronization drift simultaneously.

4. Experimental setup and methodology

This study employs a practical testbed designed to evaluate the performance and timing accuracy of LoRaWAN Class B operations under realistic conditions. The testbed implements the standard LoRaWAN architecture, comprising EDs, a GW, and a NS. The primary metric for evaluation is the synchronization precision between GW beacons and ED ping slots, which is critical for Class B performance. A critical first step was the selection of a representative ED. We conducted a comparative analysis of LoRaWAN hardware platforms (summarized in Table 3), focusing on their support for the timing requirements of Class B. The STM32 B-L072Z-LRWAN1 board was selected as the primary device for this study. This choice was motivated by its integration of a low-power STM32L072CZ MCU with the Semtech SX1276 transceiver, which provides stable and well-characterized radio performance across the 860–930 MHz range and supports Class B operations, ensuring accurate synchronization with GW beacons. In addition, the platform is widely used in academic and industrial research, as evidenced in prior studies [24], providing a benchmark for reproducibility and facilitating meaningful comparisons with previous results. While other platforms, such as the LilyGO TTGO LoRa32, offer higher computational capabilities, they do not provide the same level of standardization for low-power, Class B evaluations, making the STM32 B-L072Z-LRWAN1 more suitable for our controlled experiments.

The GW employed in this study is the Lorank8 [32], whose architecture is based on a SeedStudio BeagleBone Green [33] running the packet forwarder software. Its Radio Frequency (RF) front-end consists

Table 2

Comparison of representative hardware-based LoRaWAN studies from the related work.

Study	Class	Hardware			SF Range	Payload (B)	Consumption	Sync. Error
		Board	MCU	Transceiver				
[11]	A, C	iM880B-L	STM32L151CB	SX1272	SF7–SF11	25–115	95 mA (TX) 1.6 mA (idle) ^a 4.5 mA (TX)	—
[20]	A	Custom	MSP430FR5969	SX1276	SF7	23	0.89 μ A (sleep)	—
[22]	B	Custom	Wonder Gecko	SX1276	SF7	16	—	~11.4 μ s*
[24]	A	B-L072Z-LRWAN1	STM32L072CZ	SX1276	SF7–SF12	20	—	~13.3 ms*
This Work	B	B-L072Z-LRWAN1	STM32L072CZ	SX1276	SF10	1	6.40 mA (sleep) 16.80 mA (idle)	1.50 ms (PS8) 11.49 ms (PS64)

^a Corresponds to the *idle* mode measured for Class A.

* Synchronization error normalized to a 128 s observation window.

Table 3

Comparison of LoRaWAN hardware platforms.

Device	MCU	LoRa Transceiver	Freq. (MHz)	Tx Power (dBm)	Rx Sens. (dBm, SF12)	Class B Support
LilyGO TTGO LoRa32	ESP32-PICO-D4	SX1276/SX1278	433–923	20	–139	No
STM32WL Nucleo-64	STM32WL55JC	SX126x	433–928	22	–139	Yes
SX1272 NAMote72	STM32L152RC	SX1272	868–928	20	–137	Yes
SK-iM880B	STM32L151	SX1272	863–870	19	–138	No
SK-iM881A	STM32L081Cx	SX1272	868	14	–138	No
SAM R34 Xplained Pro-	ATSAMR34	SX1276	868/915	20	–137	No
B-L072Z-LRWAN1	STM32L072CZ	SX1276	860–930	20	–137	Yes
NUCLEO-L073RZ	STM32L073RZ	SX1261	868	14	–148	Yes
RAK4631	nRF52840	SX1262	863–928	22	–148	Yes
Murata 1SJ	STM32WL55JC	SX1262	863–928	22	–137	Yes
Seed LoRa-E5-HF	STM32WLE5JC	SX126X	868/915	22	–135	Yes

of an iC880A concentrator equipped with a Semtech SX1301 baseband, enabling the handling of multiple simultaneous LoRa channels [34]. Crucially, the integrated Neo-6M GPS module provides the precise timing reference required for Class B operation, generating a Pulse Per Second (PPS) signal with sub-microsecond accuracy for beacon synchronization [35].

On the network side, we deployed a ChirpStack NS instance. ChirpStack's modular architecture was instrumental for device management, MAC command handling, and, crucially, the scheduling and monitoring of Class B downlinks [36]. The complete physical setup of the testbed is shown in Fig. 3. It integrates the selected EDs, the Lorank8 GW with its GPS module, the logic analyzer for timing measurements, and the power monitoring unit. A hardware start button is included to trigger the simultaneous activation of the LoRaWAN communication on all EDs, ensuring a synchronized experiment initialization.

The EDs used in this work were programmed using STM32CubeIDE and the I-CUBE-LRWAN software expansion package for LoRaWAN. The GW runs the open-source packet forwarder [37]. This software forwards RF packets bidirectionally between the concentrator and NS via Internet Protocol/User Datagram Protocol (IP/UDP). The GW stack includes the *loragateway* repository [38] (HAL/driver for SX1301), updated to versions 4.0.1 and 5.0.1 for Class B support.

The evaluation of timing accuracy required precise instrumentation. We monitored the Transmission (TX) and Reception (RX) states of the ED by activating General Purpose Input/Output (GPIO) pins PA8 and PA10, respectively. Simultaneously, on the GW, we configured GPIO50 to signal the start and end of each beacon transmission. These digital signals were captured using an 8-channel AZ-Delivery logic analyzer [39], sampling at 24 Msps, to measure the temporal alignment between beacon transmissions and ping slot openings with high precision. Table 4 summarizes the GPIO connections used for measurement purposes, including beacon detection, uplink/downlink activity, and synchronization signals.

Table 4

Physical connections among testbed components.

Source Device	Pin	Destination	Destination Pin
iC880A Concentrator Connections			
iC880A	21	BeagleBone Green	9_5 (5V)
iC880A	22	BeagleBone Green	9_1 (GND)
iC880A	14	BeagleBone Green	9_22 (CLK)
iC880A	15	BeagleBone Green	9_21 (D0)
iC880A	16	BeagleBone Green	9_18 (D1)
iC880A	17	BeagleBone Green	9_17 (CS0)
iC880A	13	BeagleBone Green	9_26 (RESET)
GPS Module Connections			
GPS	VCC	BeagleBone Green	9_6 (5V)
GPS	GND	BeagleBone Green	9_2 (GND)
GPS	TX	BeagleBone Green	9_11 (RXD)
GPS	RX	BeagleBone Green	9_13 (TXD)
GPS	PPS	iC880A	Pin 19
Logic Analyzer Connections			
Logic Analyzer	CH1	BeagleBone Green	9_14 (GPIO50)
Logic Analyzer	CH2	End Device	PA8 (TX event)
Logic Analyzer	CH3	End Device	PA10 (RX event)

5. Results and discussion

This section presents the results of three complementary experiments designed to evaluate both timing and energy aspects of Class B operation on the selected ED: the B-L072Z-LRWAN1 development board (see Section 4). The experiments assess (i) synchronization precision under different ping slot densities, (ii) average current consumption in non-sleep and sleep modes, and (iii) intra-beacon timing error accumulation. Together, these experiments provide a comprehensive assessment of how ping slot configuration and hardware characteristics influence synchronization accuracy and practical energy efficiency.

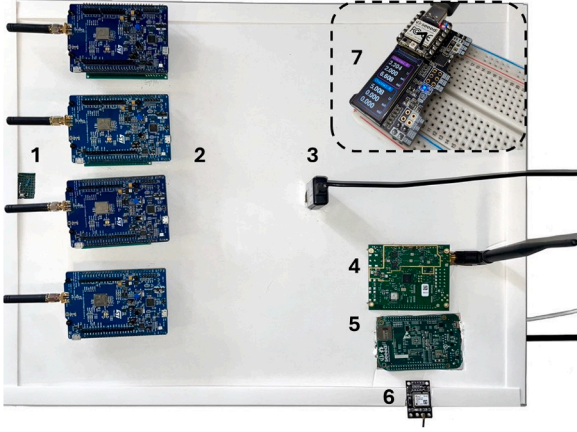


Fig. 3. Experimental testbed for Class B evaluation: (1) start button, (2) four STM32 B-L072Z-LRWAN1 EDs, (3) 8-channel logic analyzer monitoring GPIO timing signals, (4) iC880A concentrator, (5) BeagleBone Green running packet forwarder, (6) Neo-6M GPS providing PPS synchronization, and (7) XIAO PowerBread for current measurement.

Four ping slot configurations were tested (PS64, PS32, PS16, PS8). Given the fixed 128 s beacon period, these correspond to 2, 4, 8, and 16 ping slots per beacon, respectively. Each configuration was evaluated over ten consecutive beacon intervals to ensure statistically consistent timing and energy measurements.

All EDs operated under identical LoRaWAN settings: SF10, 125 kHz bandwidth, OTAA activation, ADR disabled, and regional settings compliant with REGION_EU868. The STM32L072CZ MCU [40] relied on a 16 MHz High-Speed Internal (HSI) Resistor-Capacitor (RC) oscillator, factory-calibrated to $\pm 1\%$ at 25 °C and 3.0 V.

Ambient temperature and humidity were monitored during the experiments using a SEN0227 SHT20 sensor [41] to ensure minimal variability. Temperature remained stable between 30.31 °C and 31.35 °C, and relative humidity between 41.45% and 42.29%. Tightly controlled temperatures were used because the HSI oscillator exhibits a strong temperature dependency, ensuring that the observed synchronization errors are primarily attributable to ping-slot density rather than thermal drift. In [24], clock drift measurements conducted in a temperature-controlled chamber revealed a clear and non-linear dependence on ambient temperature. Specifically, for STM32-based LoRaWAN devices, clock drift variations exceeding several hundreds of milliseconds over a 60-min interval were observed when the temperature ranged from 10 °C to 30 °C, with markedly different sensitivities across devices of the same model. These results highlight that even moderate temperature changes can significantly alter clock behavior and exacerbate inter-device heterogeneity. Consequently, restricting the temperature range in the present experiments is essential to isolate the impact of ping-slot density on synchronization accuracy and to avoid conflating it with temperature-induced clock drift.

5.1. Ping slot synchronization error analysis

In this experiment, the synchronization error across different ping slot configurations was analyzed to assess timing precision in Class B devices. Each configuration was tested through ten independent experimental runs, with timestamp data collected over ten consecutive beacon intervals per run. In the PS64 configuration, two ping slots are generated per beacon, resulting in one measured time interval between them. Similarly, PS32 produces four ping slots (three intervals), PS16 yields eight ping slots (seven intervals), and PS8 generates sixteen ping slots (fifteen intervals).

The distribution of ping slot synchronization errors is presented in Fig. 4 using violin plots. To accurately characterize the timing perfor-

Table 5

Statistical summary of synchronization error per ping slot configuration.

Ping Slot	Samples (n)	Mean Error (ms)	Std. Dev. (ms)	Coef. Var. (%)	Estimated Bias (ppm)
8	1500	1.49608	0.00606	31.10218	194.8
16	700	2.98563	0.00425	21.87951	194.4
32	300	5.80550	0.00191	10.08878	189.0
64	100	11.48555	0.00110	5.87155	186.9

mance, it is necessary to distinguish between systematic bias (clock drift) and random effects (jitter). As shown in the figure, the error distribution implies a superposition of a deterministic offset and stochastic noise.

The observed behavior can be attributed to the fact that the absolute timing jitter introduced by the MAC layer, radio stack, and measurement instrumentation remains nearly constant across all configurations, typically within the microsecond to sub-millisecond range. When ping slots are closely spaced (e.g., PS8), this fixed jitter represents a larger proportion of the inter-slot interval, increasing relative variability. Conversely, for configurations with longer intervals, the same jitter becomes a smaller fraction of the total period, resulting in a lower coefficient of variation.

Table 5 summarizes the synchronization errors. A critical observation from these results is that while the absolute mean error in milliseconds increased linearly with the ping slot period—as expected due to the longer time intervals—the mean relative error remained consistently below 0.02% across all configurations.

This behavior indicates that the dominant source of error is not random noise, but a deterministic systematic bias. By converting the relative error into Parts Per Million (ppm), we observe a stable clock drift ranging between 186 and 195 ppm across all configurations. This confirms that the error accumulates as a linear function of time, driven by the inherent inaccuracy of the HSI oscillator relative to the GW's GPS-synchronized clock.

Notably, both the standard deviation and the coefficient of variation decreased significantly as the ping slot period increased. The standard deviation decreases to as low as 1.1 μ s for PS64 (see Table 5). This sub-millisecond jitter confirms that the random component (software interruptions and processing delays) contributes minimally to the total synchronization error compared to the systematic clock drift.

Therefore, the total synchronization error can be modeled as:

$$E_{\text{total}}(t) = \underbrace{R_{\text{drift}} \cdot t}_{\text{Systematic bias}} + \underbrace{\epsilon_{\text{jitter}}}_{\text{Random noise}}$$

where $R_{\text{drift}} \approx 190$ ppm represents the oscillator offset and $\epsilon_{\text{jitter}} < 10$ μ s accounts for processing variability.

While the majority of the synchronization error is explained by this linear clock drift, the remaining random discrepancies can be attributed to secondary factors:

- **HSI oscillator inaccuracy.** The STM32L072's HSI16 internal RC oscillator has a factory-calibrated accuracy of $\pm 1\%$ [40]. Over the longest ping slot interval, corresponding to the PS64 configuration, this inaccuracy could theoretically lead to a drift of up to ± 640 ms. However, the observed errors are orders of magnitude smaller (mean of 11.49 ms for PS64), indicating that while the HSI drift is indeed the dominant error source, it behaves as a consistent systematic bias rather than a random fluctuation. The selection of the HSI oscillator over other oscillators such as TCXOs was motivated by representativeness, power efficiency, and relevance: HSI is common in cost-sensitive IoT MCUs, while using an external TCXO would increase hardware cost in many other boards; HSI consumes less energy than an external TCXO, which is important for low-power Class B applications; and characterizing HSI performance provides a baseline for typical real-world deployments where high precision oscillators are

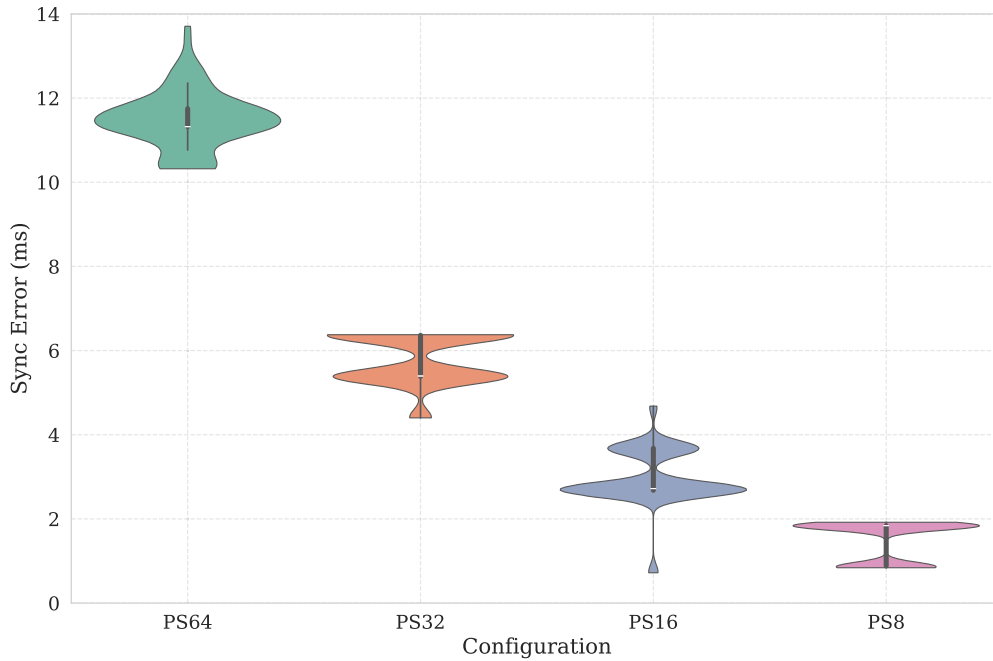


Fig. 4. Probability density of synchronization error for each ping slot configuration. Higher ping-slot densities show wider distributions, indicating greater timing variability, while lower densities yield more consistent synchronization.

not always available, even though some boards (e.g., the B-L072Z-LRWAN1) include an integrated TCXO option.

- **MAC-layer processing delays.** According to experimental evaluations of LoRaWAN stacks, tasks such as packet parsing, scheduling, and radio switching introduce non-deterministic delays ranging from tens to hundreds of microseconds [42].
- **Radio propagation delays.** Due to the finite speed of light, these delays are approximately $3.3 \mu\text{s}$ per kilometer in free-space conditions, and thus negligible for typical LoRaWAN communication ranges [43].

The combined impact of these secondary factors is at least one order of magnitude smaller than the synchronization error attributable to systematic bias accumulation over the intra-beacon interval. Consequently, the dominant factor determining synchronization precision is the linear clock drift of the HSI oscillator, while ping slot density primarily influences how this drift manifests in absolute terms.

5.2. Average current consumption estimation

Current consumption was measured using the XIAO PowerBread [44], an open-source power analyzer based on the INA3221 sensor.¹ Data was sampled at 10 ms intervals to capture high-resolution temporal profiles of the current drawn across different communication stages. All current measurements reflect the total power consumption of the B-L072Z-LRWAN1 Discovery board, including the MCU, LoRa module (SX1276), Real-Time Clock (RTC), system clocks (SYSCLK, HCLK, PCLK), Serial Peripheral Interface (SPI) for communication with the LoRa module, and onboard voltage regulators. These values therefore represent system-level power draw under realistic operating conditions, not isolated MCU consumption. Figs. 5 and 6 illustrate typical current profiles for Always-on and Sleep mode, respectively.

The average and peak currents for each communication stage are summarized in Table 6.

To isolate the impact of the ping slot configuration on energy consumption, the current contributions of periodic Class B events were calculated as:

$$I_{\text{event}} = \frac{I_{\text{stage}} \cdot t_{\text{stage}}}{T_{\text{event}}},$$

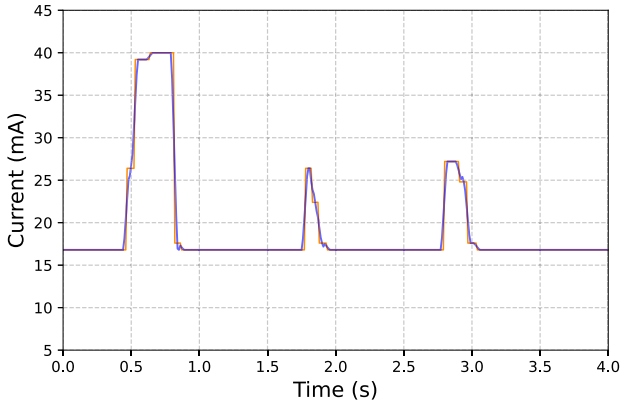
where I_{stage} and t_{stage} are taken from Table 6, and T_{event} is the period of the corresponding activity ($T_{\text{uplink}} = 600 \text{ s}$, $T_{\text{beacon}} = 128 \text{ s}$, T_{ping} varies with the ping slot configuration).

The results in Table 7 confirm that the ping slot activity dominates the variable energy consumption. As the ping slot period decreases, the number of reception windows per unit time increases, effectively doubling the ping-related current contribution when period is halved. Sleep mode reduces overall average current by approximately 62% compared to Always-on, demonstrating substantial energy efficiency improvement.

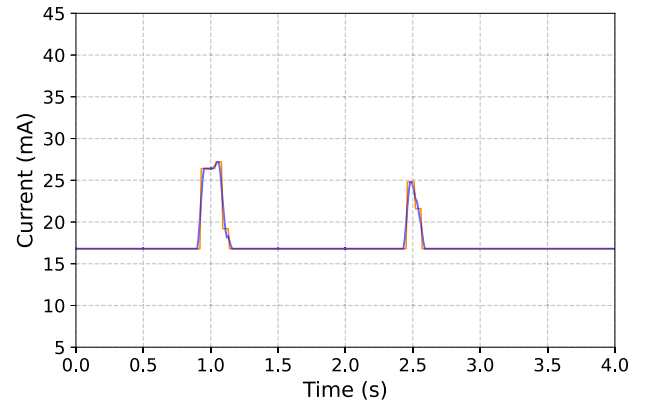
When these energy measurements are considered alongside the synchronization accuracy analysis presented in Section 5.3 and Table 5, a clear trade-off emerges between timing precision and energy efficiency. Configurations with shorter ping slot periods (e.g., PS8) achieve slightly lower absolute synchronization errors at the expense of significantly higher current consumption. Conversely, PS64 yields the lowest average current —16.92 mA Always-on, 6.45 mA Sleep— while maintaining absolute timing errors below 12.8 ms (within typical 25 ms receive window) and relative errors below 0.02%.

For a 1000 mAh battery, these consumption levels correspond to estimated operational lifetimes of approximately 2.46 days in Always-on mode and 6.46 days in Sleep mode. These lifetime estimates reflect the performance of a general-purpose development board and are intended for comparative evaluation between Class B configurations, rather than representing the battery lifetime achievable with application-optimized hardware and software. Therefore, PS64 represents the most balanced configuration, providing predictable synchronization performance with minimal energy overhead. Further optimization could be achieved by leveraging the STM32L0's STOP mode [45], which offers additional current savings beyond those measured in Sleep mode.

¹ <https://www.ti.com/product/INA3221>

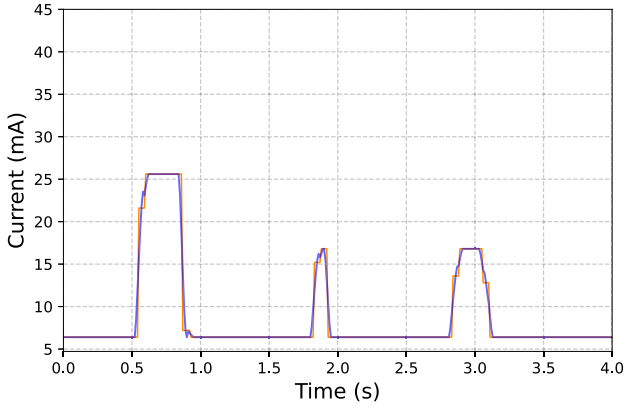


(a) Uplink, RX1 and RX2

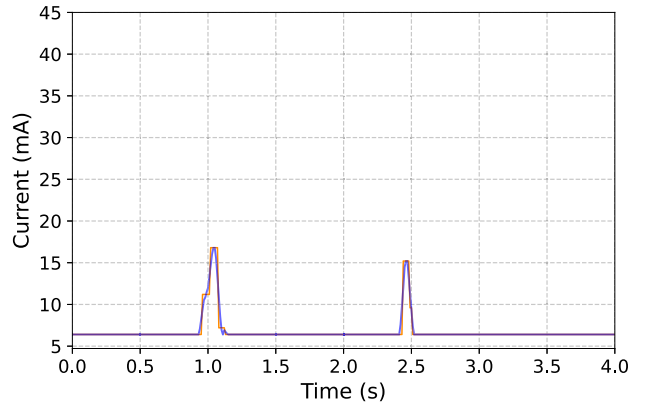


(b) Beacon and Ping Slot

Fig. 5. Current consumption profiles for Class B operations in Always-on mode: (a) uplink transmission with subsequent RX1 and RX2 reception windows, and (b) beacon followed by a ping slot event.



(a) Uplink, RX1 and RX2



(b) Beacon and Ping Slot

Fig. 6. Current consumption profiles for different Class B operations in Sleep mode: (a) uplink transmission with subsequent RX1 and RX2 reception windows, and (b) beacon followed by a ping slot event. The ED enters sleep mode between communication events, reducing the idle baseline from 16.8 mA (Always-on) to 6.4 mA.

Table 6

Comparison of average, peak, and duration per communication stage between Always-on and Sleep modes.

Stage	Always-on			Sleep Mode		
	Average (mA)	Peak (mA)	Duration (ms)	Average (mA)	Peak (mA)	Duration (ms)
Uplink	34.81	40.00	390 ^a	25.00	25.60	330 ^a
RX1	21.85	26.40	160	15.87	16.80	100
RX2	23.80	27.20	240	14.93	16.80	280
Beacon	24.27	27.20	200	15.87	16.80	100
Ping Slot	23.35	24.80	110 ^b	12.80	15.20	70 ^b
Idle	16.80	16.80	Continuous	6.40	6.40	Continuous

^a Measured uplink durations (390 ms in Always-on mode, 330 ms in Sleep mode) exceed the theoretical Time-on-Air of 288.8 ms for a 1-byte payload with SF10, BW 125 kHz, CR 4/5, and an 8-symbol preamble in EU868.

^b Measured Ping Slot durations are 110 ms in Always-on mode and 70 ms in Sleep mode, both differing from the theoretical 30 ms specified in the LoRaWAN standard [1].

5.3. Timing aggregation experiment

To evaluate how timing errors propagate and accumulate within a single beacon period, a timing aggregation experiment was conducted using four B-L072Z-LRWAN1 Discovery kits equipped with Murata CMWX1ZZABZ-091 modules [46]. The devices comprised two manufacturing batches: ED1 and ED2 (2018, inspection SS8516005) and ED3 and ED4 (2017, inspection SS7N25009). Prior characterization of these

devices revealed significant clock skew variations between batches [24], providing context for analyzing temporal drift patterns.

The cumulative timing error at the k th ping slot within a beacon period, denoted $\Delta_{\text{agg}}^{(k)}$, was calculated as the sum of individual timing errors:

$$\Delta_{\text{agg}}^{(k)} = \sum_{i=1}^k (T_{\text{slot}} - (t_{i+1} - t_i))$$

Table 7
Current contributions of periodic Class B events for different PS periods.

Ping Slot	Always-on (mA)				Sleep mode (mA)			
	Uplink + RX	Beacon	Ping Slot	Total I_{avg}	Uplink + RX	Beacon	Ping Slot	Total I_{avg}
8	0.04	0.04	0.32	17.20	0.02	0.01	0.11	6.54
16	0.04	0.04	0.16	17.04	0.02	0.01	0.06	6.49
32	0.04	0.04	0.08	16.96	0.02	0.01	0.03	6.46
64	0.04	0.04	0.04	16.92	0.02	0.01	0.01	6.45

Notes: Total I_{avg} includes idle current of 16.80 mA for Always-on and 6.40 mA for Sleep mode. Values are rounded to two decimal places.

Table 8
Quantitative NTN timing and Doppler budget vs. measured hardware performance.

Parameter	LEO (600 km)	GEO	Impact of Measured 190 ppm Drift
Avg. Propagation Delay	~ 2–10 ms	~ 240–280 ms	Requires pre-emptive transmission timing Hardware drift consumes 24.3 ms, leaving no margin for dynamic jitter absorption
Delay Variation (Jitter)	~ 2–8 ms	< 1 ms	Severe sub-carrier/channel misalignment
Max Doppler Shift (868 MHz)	±15 kHz	Negligible	Signal shifts outside the 125 kHz BW
Clock-induced Frequency Offset	≈ 165 kHz	≈ 165 kHz	

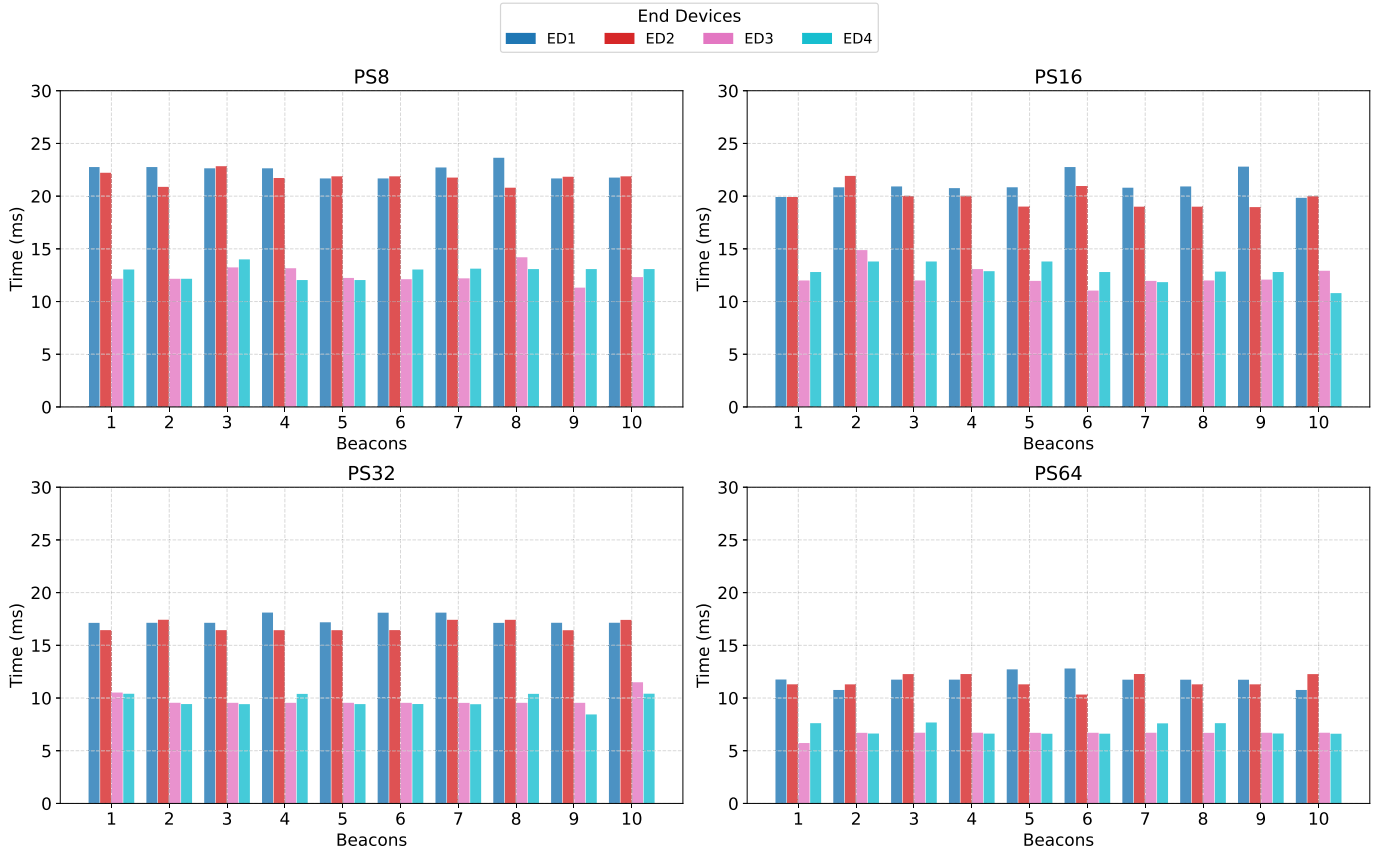


Fig. 7. Cumulative ping slot timing deviations (Δ_{agg}) across ten consecutive beacon periods for four EDs under configurations of 8, 16, 32, and 64 ping slots. Each plot shows a linear intra-beacon accumulation that resets at the start of every new beacon.

where $T_{slot} = 122.880/N$ seconds represents the ideal slot duration, N is the number of ping slots per beacon period, and $(t_{i+1} - t_i)$ is the measured interval between consecutive slots.

Fig. 7 shows Δ_{agg} across ten beacon periods. Devices from the same manufacturing batch (ED1, ED2) exhibit nearly identical drift patterns, while the older batch (ED3, ED4) shows slightly higher accuracy. This suggests batch-specific factors—particularly oscillator calibration—influence timing performance more than device age.

These findings show that synchronization errors accumulate predictably within each beacon period, reaching their maximum at the last ping slot before the next beacon is received. Although cyclic behavior prevents unbounded error accumulation, drift magnitude within a single period is critical for downlink reliability. To assess operational robustness, we compare the worst-case cumulative offsets observed across all devices and beacon intervals against the nominal receive window defined by the LoRaWAN specification. The LoRaWAN specification de-

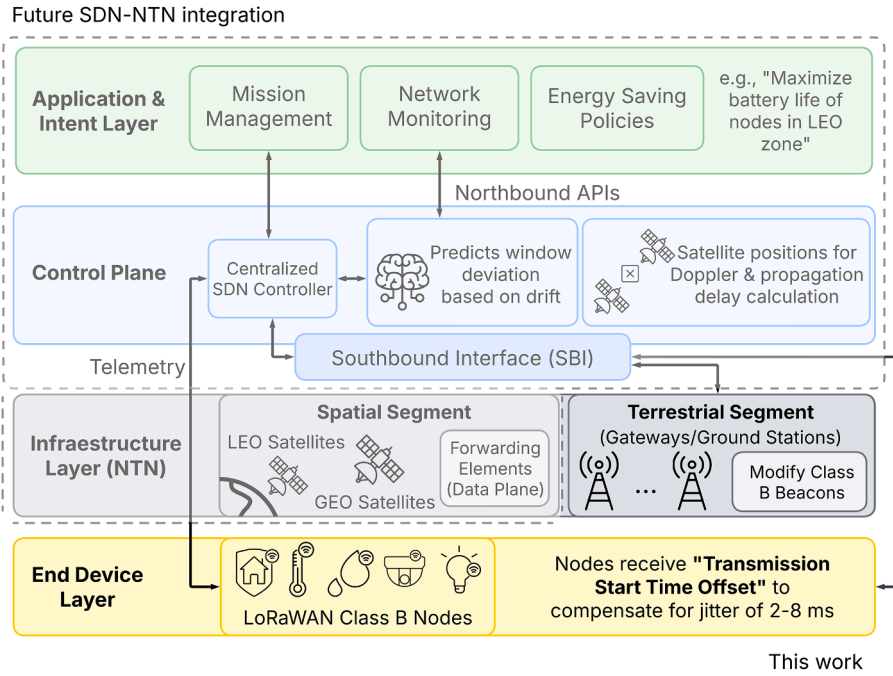


Fig. 8. Conceptual SDN-based architecture illustrating NTN and IoT integration.

finds a nominal ping slot duration of 30 ms [1], which represents the full time window during which a GW may transmit and an ED is expected to listen.

In our experiments, the maximum cumulative drift was 23.6 ms (observed on ED1 during Beacon 8 under PS8 configuration), occurring at the end of the beacon period. This corresponds to 78.7% of the 30 ms slot duration, leaving a safety margin of only 6.4 ms before the device's reception window would fall outside the nominal slot. In contrast, the worst-case drift for PS64 was 12.8 ms (ED1, Beacon 6), consuming only 42.7% of the 30 ms budget and preserving a much more robust safety margin of 17.2 ms.

These results indicate that, while configurations such as PS8 remain operationally feasible in stable terrestrial conditions, they operate near the synchronization limit. Minor perturbations—such as temperature shifts beyond our test range, supply voltage variations, or oscillator aging—could easily consume the remaining 6.4 ms margin, leading to downlink loss. Consequently, configurations like PS32 or PS64 provide a more stable balance for reliable deployments.

5.4. Timing constraints for NTN integration and SDN-based orchestration

Characterizing this internal timing behavior is a mandatory prerequisite for integrating Class B devices into NTNs. While terrestrial gateways are static, LEO satellites at 600 km altitude move at an angular velocity of approximately 1.09×10^{-3} rad/s, as derived from Kepler's third law. The average propagation delay for such a satellite ranges from approximately 2 ms at zenith to 10 ms at low elevation angles. More critically for synchronization, this delay varies dynamically by ~ 2 –8 ms during a single pass [47], a range that directly impacts the timing budget. When mapping our measured hardware drift ($R_{\text{drift}} \approx 190$ ppm) to these dynamics, the vulnerability of the synchronization budget becomes evident. Over a 128 s beacon period, a 190 ppm drift accumulates ≈ 24.3 ms of error. In a LEO scenario, this hardware-induced error consumes nearly the entire 30 ms LoRaWAN window, leaving virtually no margin to accommodate the ~ 2 –8 ms of satellite-induced delay jitter.

As summarized in Table 8, the timing constraints differ significantly between orbital shells. While Geostationary Earth Orbit (GEO) satellites exhibit much higher absolute delays, their stationary nature relative to

the ED results in negligible delay jitter, making them more tolerant to the measured hardware drift. Conversely, in LEO, the 190 ppm instability not only threatens timing alignment but also exacerbates Doppler-induced frequency shifts (up to ± 15 kHz at 868 MHz [48]). For narrow-band modulations like LR-FHSS, an uncompensated 190 ppm drift adds a ≈ 165 kHz offset. This offset exceeds the typical 125 kHz bandwidth of standard LoRa channels, effectively shifting the signal out of the intended frequency grid and rendering standard receiver synchronization mechanisms ineffective. These findings confirm that the HSI oscillator acts as a performance bottleneck for NTN-integrated IoT, underscoring the necessity for either TCXO-based hardware or mechanisms to adjust the transmission start time to compensate for dynamic channel delays.

The orchestration of dynamic timing adjustments and frequency compensations in heterogeneous NTN environments requires a flexible and centralized management paradigm. In this context, SDN has emerged as a promising approach to manage constrained IoT infrastructures by decoupling the control plane from the data plane and enabling global network visibility and programmability [31].

Several SDN-based architectures have been proposed for IoT networks operating over LLNs, such as SDN-WISE, TinySDN, and CoralSDN, each targeting different challenges including routing optimization, mobility management, and energy efficiency. These architectures leverage centralized controllers to dynamically configure network parameters, adapt routing behavior, and monitor network conditions, which is particularly beneficial in highly dynamic or resource-constrained environments [31].

In the context of NTN-IoT integration, a similar SDN-based approach could facilitate the coordination of timing and synchronization mechanisms required by satellite links. As illustrated in Fig. 8, such an architecture may follow a multi-layer SDN structure where the infrastructure and data planes execute packet forwarding and basic synchronization tasks, while the control plane dynamically adjusts transmission parameters based on network conditions.

To enable this functionality, the control layer would require mechanisms capable of estimating two key factors affecting communication reliability in NTN scenarios: (i) the temporal drift introduced by hardware clock inaccuracies and propagation delays, and (ii) the relative position and movement of satellites, which influence link availability and

timing windows. These estimations could be used by the controller to adapt transmission schedules or compensate timing offsets while avoiding excessive signaling overhead over satellite links.

At the top of the architecture, the application and intent layer would provide high-level network management functions such as mission configuration, monitoring, or energy optimization policies. Through northbound interfaces, these objectives could be translated into control decisions that dynamically adapt network behavior while keeping the complexity away from constrained IoT devices.

6. Conclusions

This study provides a comprehensive experimental analysis of the timing-energy trade-offs in Class B, revealing how synchronization accuracy and power consumption are jointly shaped by protocol configuration and hardware characteristics.

The primary finding of this work is the identification of a predictable, cyclic pattern of timing error accumulation within each beacon period. Synchronization error grows linearly after each beacon reception, reaching its maximum just before the next beacon. This accumulation—driven by ping slot density and hardware oscillator characteristics—is the dominant source of absolute error in longer ping slot periods (e.g., PS64). Crucially, the error resets with every beacon, preventing long-term drift.

Our experiments confirm a clear trade-off between synchronization precision and energy efficiency. Higher ping slot densities (e.g., PS8) yield lower absolute synchronization errors (~1.5 ms) but consume more energy, whereas lower densities (e.g., PS64) minimize energy usage at the cost of higher absolute errors (~11.5 ms). However, even the PS64 error remains well within a typical 25 ms receive window, demonstrating operational viability for applications that do not require millisecond-level precision.

From an energy perspective, implementing low-power sleep modes is the most significant factor for battery lifetime, reducing average system current by approximately 62%. This hardware-level optimization outweighs marginal gains from ping slot parameter tuning. The timing aggregation experiment revealed that devices from the same manufacturing batch exhibit nearly identical drift profiles, highlighting oscillator calibration as a key hardware-dependent factor for multi-device synchronization.

For time-critical applications, such as industrial control or coordinated sensing, configurations like PS8 or PS16 are recommended, using hardware with high-stability oscillators from consistent manufacturing batches. For energy-constrained monitoring, such as environmental or agricultural sensing, PS32 or PS64 combined with aggressive sleep modes provide extended lifetime while maintaining robust timing performance. In large-scale heterogeneous networks, a mixed approach is beneficial, assigning dense ping slots to latency-sensitive nodes and sparse configurations to monitoring devices, thereby optimizing overall network efficiency.

This work also opens several avenues for future research. The predictable nature of intra-beacon error suggests that software-based compensation algorithms could correct drift in real time. Adaptive ping slot scheduling, dynamically adjusting density based on application requirements and remaining battery capacity, represents another opportunity for autonomous network optimization.

Beyond device-level optimizations, our experimental characterization of 190 ppm hardware drift indicates that synchronization in NTN cannot be treated as a purely local process. When projected to a LEO environment with dynamic propagation delays of ~2–8 ms, the available synchronization budget becomes highly constrained.

These findings underscore the necessity of network-level coordination, which jointly accounts for hardware-induced frequency offsets, satellite mobility, and varying propagation delays when scheduling transmissions. Centralized control paradigms, inspired by SDN architectures for IoT and LPWANs, are particularly suitable in this context,

providing global network visibility and enabling dynamic adaptation of network behavior.

Rather than relying solely on local device adjustments, such coordinated approaches can orchestrate both synchronization and transmission scheduling across the network, enhancing scalability and robustness in hybrid terrestrial and non-terrestrial IoT infrastructures. Extending Class B to NTN scenarios is especially promising, as it addresses the challenges posed by highly dynamic channel conditions. Furthermore, recent regulatory developments, such as Electronic Communications Committee (ECC) approval for satellite-to-device communications in the 862–870 MHz Short-Range Device (SRD) band in Europe [49], open new opportunities for NTN-based LoRaWAN deployments. Our terrestrial testbed and detailed timing characterization thus provide a valuable reference for designing future NTN-LoRaWAN architectures.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to review and improve the English grammar and fluency of the manuscript. After using this tool, the authors carefully reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

María Ángeles Amador: Writing – original draft, Visualization, Software, Investigation, Data curation; **Celia Garrido-Hidalgo:** Writing – review & editing, Supervision, Methodology, Conceptualization; **Luis Roda-Sanchez:** Writing – review & editing, Supervision, Methodology, Conceptualization; **Elena Garrido-Martínez:** Methodology, Investigation; **Teresa Olivares:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Maria Angeles Amador reports financial support was provided by University of Castilla-La Mancha. Celia Garrido-Hidalgo reports financial support was provided by University of Castilla-La Mancha. Luis Roda-Sanchez reports financial support was provided by University of Castilla-La Mancha. Elena Garrido-Martinez reports financial support was provided by University of Castilla-La Mancha. Teresa Olivares reports financial support was provided by University of Castilla-La Mancha. Maria Angeles Amador reports a relationship with University of Castilla-La Mancha that includes: employment. Celia Garrido-Hidalgo reports a relationship with University of Castilla-La Mancha that includes: employment. Luis Roda-Sanchez reports a relationship with University of Castilla-La Mancha that includes: employment. Elena Garrido-Martinez reports a relationship with University of Castilla-La Mancha that includes: employment. Teresa Olivares reports a relationship with University of Castilla-La Mancha that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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